

HAT-P-36b

R-0915

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230520 · created 2026-07-29T07:19:25+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460084.8065 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.044
Transit depth (Rp/Rs)^2	1.83 +/- 1.18 [%]
Orbital Inclination (inc)	83.14 +/- 2.81
Airmass coefficient 1 (a1)	0.99 +/- 0.019
Airmass coefficient 2 (a2)	0.007 +/- 0.03
Scatter in the residuals of the lightcurve fit is	1.91 %
Best Comparison Star	#3 - [287, 368]
Optimal Aperture	4.35
Optimal Annulus	20.65
Transit Duration (day)	0.084 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.135 ± 0.044 vs 0.126 (deviation 0.2σ)
Duration fit vs published	0.084 d vs 0.0925 d (deviation 0.61σ)
Mid-time O–C	12.5 min
Depth SNR	3.1
Residual scatter	1.91 %

Light curve

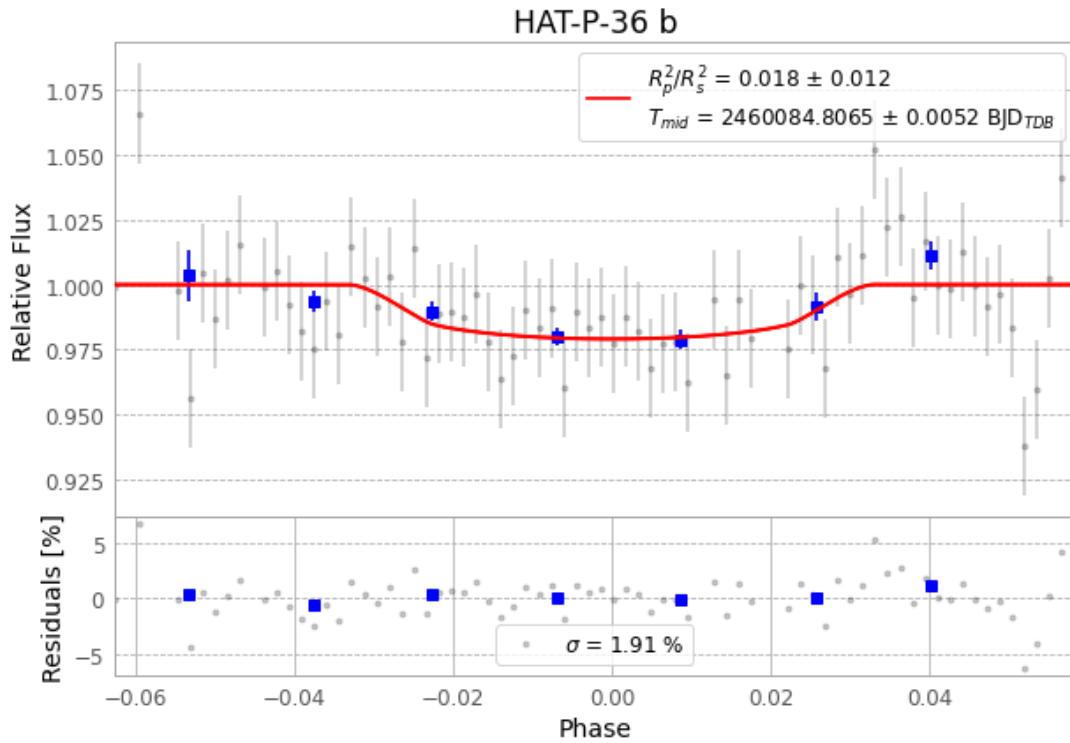


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-05-19.png (SHA-256 59899b7f51caadf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 384], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 29e9532c25cca7ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

82 FITS frames (54 MB), HATP-36230520050617.FITS → HATP-36230520090915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230520.log (b58190f71c19...), FinalLightCurve_HAT-P-36 b_2023-05-19.png (59899b7f51ca...), FinalLightCurve_HAT-P-36 b_2023-05-19.csv (d90a200e9b0f...)

Compute resources

Wall time 230.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 07:19Z.